



TCXO / VC-TCXO

HIGH STABILITY / Low noise



Product Number
TG2016SMN : X1G005441xxxx25
TG2520SMN : X1G005421xxxx27

TG2016SMN / TG2520SMN

- Output frequency : 10 MHz to 55 MHz
- Supply voltage : 1.8 V Typ./ 2.8 V Typ./ 3.0 V Typ./ 3.3 V Typ.
- Frequency / temperature characteristics
 - : $\pm 0.5 \times 10^{-6}$ Max. (-40 °C to +85 °C)
 - : $\pm 2.0 \times 10^{-6}$ Max. (-40 °C to +85 °C)
- External dimensions: 2.0 × 1.6 × 0.73 mm / 2.5 × 2.0 × 0.8 mm
- Applications : GPS, RF
Wireless communication devices
(LTE, WiMAX, Wi-Fi, W-LAN, IoT other)
- Features : Low noise



TG2016SMN
(2.0 × 1.6 × 0.73 mm)



TG2520SMN
(2.5 × 2.0 × 0.8 mm)

Specifications (characteristics)

Item	Symbol	TCXO	VC-TCXO	Conditions / Remarks
Output frequency range	f _o	10 MHz to 55MHz 16, 16.368, 16.369, 19.2, 20, 24, 25, 26, 27, 27.6, 30, 32, 38.4, 40, 48, 50, 52 MHz		Standard frequency
Supply voltage	V _{CC}	1.8 V ± 0.1 V / 2.8 V ± 5 % / 3.0 V ± 5 % / 3.3 V ± 5 %		Supply voltage range : 1.7 V to 3.63 V
Storage temperature range	T _{stg}	-40 °C to +90 °C		Storage as single product.
Operating temperature range	T _{use}	G: -40 °C to +85 °C		
Frequency tolerance	f _{tol}	±1.5 × 10 ⁻⁶ Max.		After reflow, +25 °C
Frequency/temperature characteristics	f _o -T _C	C: ±0.5 × 10 ⁻⁶ Max. / -40 °C to +85 °C F: ±2.0 × 10 ⁻⁶ Max. / -40 °C to +85 °C		Standard stability version
Frequency/load coefficient	f _o -Load	±0.1 × 10 ⁻⁶ Max.		10 kΩ // 10 pF ± 10 %
Frequency/voltage coefficient	f _o -V _{CC}	±0.1 × 10 ⁻⁶ Max.		V _{CC} ± 5 %
Frequency aging	f _{age}	±0.5 × 10 ⁻⁶ Max.		+25 °C, First year, f _o = 10MHz, 12 MHz ≤ f _o ≤ 20 MHz, 24 MHz ≤ f _o ≤ 40 MHz
		±1.5 × 10 ⁻⁶ Max.		+25 °C ,First year, 10 MHz < f _o < 12 MHz, 20 MHz < f _o < 24 MHz, 40 MHz < f _o ≤ 55 MHz
Current consumption	I _{CC}	1.5 mA Max.		10 MHz ≤ f _o ≤ 26 MHz
		1.8 mA Max.		26 MHz < f _o ≤ 40 MHz
		2.0 mA Max.		40 MHz < f _o ≤ 50 MHz
		2.1 mA Max.		50 MHz < f _o ≤ 55 MHz
Input impedance	Z _{in}	500 kΩ Min.	-	V _C - GND (DC)
Frequency control range	f _{cont}	-	±5.0 × 10 ⁻⁶ Min.	B: V _C = 0.9 V ± 0.6 V (V _{CC} = 1.8 V) or C: V _C = 1.4 V ± 1.0 V (V _{CC} = 2.8 V) or D: V _C = 1.5 V ± 1.0 V (V _{CC} = 3.0 V) or E: V _C = 1.65 V ± 1.0 V (V _{CC} = 3.3 V)
Frequency change polarity	f _{cp}	-	Positive polarity	
Symmetry	SYM	40 % to 60 %		GND level (DC cut)
Output voltage	V _{pp}	0.8 V Min.		Peak to Peak
Start-up time	t _{str}	1.0 ms Max.		t = 0 at 90% V _{CC}
Output load	Load_R	10 kΩ		DC cut capacitor = 0.01 μF
	Load_C	10 pF		

* Note : Please contact us for requirements not listed in this specification.

Product Name **TG2016 SMN 26.000000MHz** **E C G N N M**
 (Standard form) ① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

① Model (TG2016, TG2520)

② Output (S: Clipped sine wave) ③ Frequency

④ Supply voltage (Refer to symbol table) ⑤ Frequency / temperature characteristics (C: ±0.5 × 10⁻⁶ Max., F: ±2.0 × 10⁻⁶ Max.)

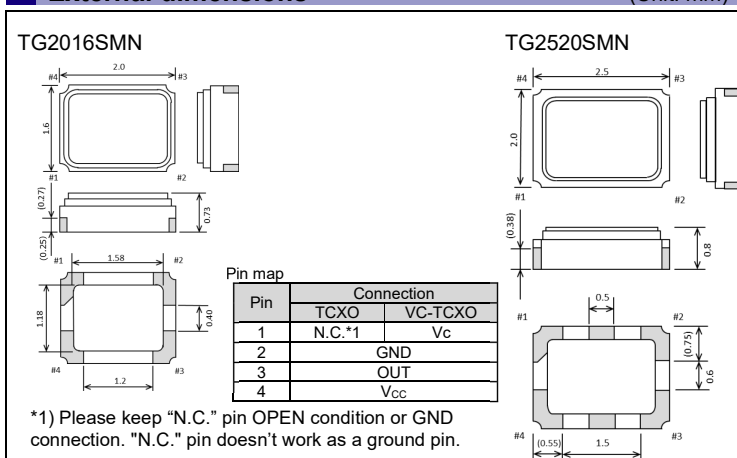
⑥ Operating temperature (G: -40 °C to +85 °C) ⑦ ST function (N: Non)

⑧ V_C function (Refer to symbol table, A: V_C = any) ⑨ Internal identification code ("M" is default)

④ Supply voltage[V _{CC}], ⑧ V _C function[V _C] (Symbol table)					
Voltage [V]	TCXO	VC-TCXO			
④ V _{CC} (Typ.)	E:1.8 M:2.8 to 3.3	E:1.8	B:2.8	A:3.0	C:3.3
⑧ V _C (Typ.)	N: Non	B 0.9	C:1.4	D 1.5	E 1.65

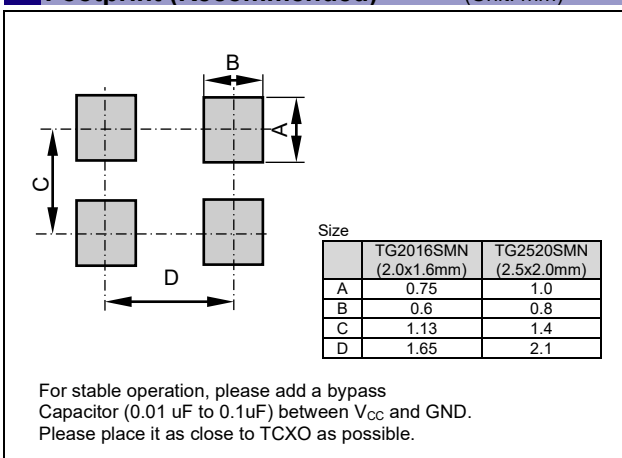
External dimensions

(Unit: mm)



Footprint (Recommended)

(Unit: mm)



PROMOTION OF ENVIRONMENTAL MANAGEMENT SYSTEM CONFORMING TO INTERNATIONAL STANDARDS

At Seiko Epson, all environmental initiatives operate under the Plan-Do-Check-Action (PDCA) cycle designed to achieve continuous improvements. The environmental management system (EMS) operates under the ISO 14001 environmental management standard.

All of our major manufacturing and non-manufacturing sites, in Japan and overseas, completed the acquisition of ISO 14001 certification.

ISO 14000 is an international standard for environmental management that was established by the International Standards Organization in 1996 against the background of growing concern regarding global warming, destruction of the ozone layer, and global deforestation.

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In order provide high quality and reliable products and services than meet customer needs, Seiko Epson made early efforts towards obtaining ISO9000 series certification and has acquired ISO9001 for all business establishments in Japan and abroad. We have also acquired IATF 16949 certification that is requested strongly by major automotive manufacturers as standard.

IATF 16949 is the international standard that added the sector-specific supplemental requirements for automotive industry based on ISO9001.

► Explanation of the mark that are using it for the catalog

	► Pb free.
	► Complies with EU RoHS directive. *About the products without the Pb-free mark. Contains Pb in products exempted by EU RoHS directive. (Contains Pb in sealing glass, high melting temperature type solder or other.)
	► Designed for automotive applications such as Car Multimedia, Body Electronics, Remote Keyless Entry etc.
	► Designed for automotive applications related to driving safety (Engine Control Unit, Air Bag, ESC etc).

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TG2520SMN 26.0000M-TCGNNM0	TG2520SMN 26.0000M-TCGNNMB	TG2520SMN 16.0000M-ECGNNM0
TG2520SMN 16.0000M-ECGNNM5	TG2520SMN 16.0000M-ECGNNMB	TG2520SMN 16.0000M-MCGNNM0
TG2520SMN 16.0000M-MCGNNM5	TG2520SMN 16.0000M-MCGNNMB	TG2520SMN 16.3680M-ECGNNM0
TG2520SMN 16.3680M-ECGNNM5	TG2520SMN 16.3690M-ECGNNM0	TG2520SMN 16.3690M-ECGNNM5
TG2520SMN 16.3690M-ECGNNMB	TG2520SMN 16.3690M-MCGNNM0	TG2520SMN 16.3690M-MCGNNM5
TG2520SMN 16.3690M-MCGNNMB	TG2520SMN 19.2000M-ACGNDM0	TG2520SMN 19.2000M-ACGNDM3
TG2520SMN 19.2000M-ACGNDM5	TG2520SMN 19.2000M-ACGNDMB	TG2520SMN 19.2000M-BCGNCM0
TG2520SMN 19.2000M-BCGNCMB	TG2520SMN 19.2000M-ECGNNM0	TG2520SMN 19.2000M-ECGNNM5
TG2520SMN 19.2000M-MCGNNM0	TG2520SMN 19.2000M-MCGNNM5	TG2520SMN 20.0000M-ECGNNM0
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TG2520SMN 24.0000M-ECGNNM5	TG2520SMN 24.0000M-ECGNNMB	TG2520SMN 24.0000M-MCGNNM0
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TG2520SMN 25.0000M-MCGNNMB	TG2520SMN 25.0000M-TCGNNMB	TG2520SMN 26.0000M-BFGNCM0
TG2520SMN 26.0000M-BFGNCM3	TG2520SMN 26.0000M-BFGNCM5	TG2520SMN 26.0000M-BFGNCMB
TG2520SMN 26.0000M-ECGNBMB	TG2520SMN 26.0000M-ECGNNM5	TG2520SMN 26.0000M-MCGNNM0
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TG2520SMN 27.6000M-MCGNNM5	TG2520SMN 30.0000M-ECGNNM0	TG2520SMN 30.0000M-ECGNNM5
TG2520SMN 32.0000M-ECGNNM0	TG2520SMN 32.0000M-ECGNNM5	TG2520SMN 32.0000M-MCGNNM0
TG2520SMN 32.0000M-MCGNNM5	TG2520SMN 32.0000M-TCGNNM0	TG2520SMN 32.0000M-TCGNNM3
TG2520SMN 32.0000M-TCGNNMB	TG2520SMN 38.4000M-ECGNBM0	TG2520SMN 38.4000M-ECGNBM3
TG2520SMN 38.4000M-ECGNBM5	TG2520SMN 38.4000M-ECGNBMX	TG2520SMN 38.4000M-ECGNNM0
TG2520SMN 38.4000M-MCGNNM0	TG2520SMN 40.0000M-CCGNDM0	TG2520SMN 40.0000M-CCGNDM3

<u>TG2520SMN 40.0000M-CCGNM5</u>	<u>TG2520SMN 40.0000M-CCGNMB</u>	<u>TG2520SMN 40.0000M-ECGNBMB</u>
<u>TG2520SMN 40.0000M-ECGNNM0</u>	<u>TG2520SMN 40.0000M-ECGNNM5</u>	<u>TG2520SMN 40.0000M-MCGNNM0</u>
<u>TG2520SMN 40.0000M-MCGNNM5</u>	<u>TG2520SMN 40.0000M-MCGNNMB</u>	<u>TG2520SMN 45.0000M-TCGNNHB</u>
<u>TG2520SMN 46.0000M-TCGNNHB</u>	<u>TG2520SMN 48.0000M-ECGNNM0</u>	<u>TG2520SMN 48.0000M-ECGNNM5</u>
<u>TG2520SMN 48.0000M-MCGNNM0</u>	<u>TG2520SMN 48.0000M-MCGNNM5</u>	<u>TG2520SMN 50.0000M-ACGNM0</u>
<u>TG2520SMN 50.0000M-ACGNMB</u>	<u>TG2520SMN 50.0000M-ECGNNM0</u>	<u>TG2520SMN 50.0000M-ECGNNM5</u>
<u>TG2520SMN 50.0000M-ECGNNMB</u>	<u>TG2520SMN 50.0000M-MCGNNM0</u>	<u>TG2520SMN 50.0000M-MCGNNM5</u>
<u>TG2520SMN 50.0000M-MCGNNMB</u>		